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DS 4002

Using Data Science to Explore Public Sentiment about COVID-19 Announcements

The COVID-19 pandemic in the United States was a completely unprecedented time that majorly impacted our society. In particular, the pandemic had large impacts on the mental, physical, and economic health of society. With little knowledge about what the timeline would be for a vaccine or treatment to stop the spread of COVID, the United States government began to implement major decisions in order to stop the spread of the virus. Several U.S. states announced major shutdowns on March 15th, 2020, which triggered large uncertainty socially and economically, as this had never previously happened in any of our lifetimes. Given the limited amounts of in person interactions, social media use increased, and many used various platforms to express their feelings about COVID. Twitter, in particular, was a key platform for these discussions.

There are existing studies about the general trends in COVID-19 discussions, but there is limited research on how sentiment evolved between the initial announcement of shutdowns and the subsequent weeks as people adjusted to new realities. An understanding of the sentiment following these shutdowns would give critical insights into these public health crisis measures taken. The analysis of tweets ranging from March to April of 2020 could lead to these important insights of public sentiment following these initial shutdowns. A data scientist like you would be able to use available datasets to draw critical conclusions about how society responded to these shutdowns and have an increased understanding of the impacts of these major decisions.

The Deliverable

Now that the height of COVID-19 is behind us, the United States government and health officials are looking for a data scientist to analyze text data from tweets taken from the March to April of 2020 timeframe and perform a sentiment analysis on this data. Given that there has not been an analysis of the public sentiment before and after major shutdown announcements, these stakeholders find it important to have a better grasp on how our society reacted to these decisions. While the hope is that we never have to experience a pandemic again in our lifetimes, it is important that we reflect on the past to help educate future decision making. In preparation for the final deliverable, an initial exploration of the dataset is necessary, as well as any potential data cleaning necessary. The deliverable will use a VADER sentiment analysis, with important statistics and graphs portrayed in an easily digestible format for policymakers and officials to be able to draw insights from.